

Data, Papers, Code on Trash Collection (AISIN Europe)

Papers and Repos

- [GitHub - majsylw/litter-detection-review: A list of useful resources in the trash classification and detection \(mainly plastic\), such as datasets, papers, links to open source projects](#)
- [Evaluation of vision transformers for the detection of fullness of garbage bins for efficient waste management](#)
- [Real time intelligent garbage monitoring and efficient collection using YOLOv8 and YOLOv5 deep learning models for environmental sustainability | Scientific Reports](#)
- [Municipal solid waste classification and real-time detection using deep learning methods](#)
- [Trash detection on the floor with Intel RealSense](#)
- [Trash Image Classification System using Machine Learning and Deep Learning Algorithms](#)

Datasets

- [Outdoor Garbage Dataset | 5000 Photos — Unidata](#)
- [GitHub - AgaMiko/waste-datasets-review: List of image datasets with any kind of litter, garbage, waste and trash](#)

Closest to our Project:

- [GitHub - julianofernandez/Litter-Detection: I am hoping to use car mounted camera data to map out trash levels in an urban environment.](#)

Open Source Models:

Model	GitHub Link	License	Notes
RF-DETR	RF-DETR GitHub (Roboflow)	Apache-2.0	Real-time transformer detector + segmentation
RT-DETR	RT-DETR Official GitHub	Apache-2.0	Real-time DETR family
Co-DETR	Sense-X/Co-DETR GitHub	MIT	SOTA DETR variant
RTMDet	OpenMMLab RTMDet via MMDetection	Apache-2.0	CNN-based real-time detector

Objectives that we could attend by classifying images that contain waste (garbage):

- **Smart Bin Monitoring:** Using local cameras to track in real-time whether public bins are full, empty, or overflowing.
- **Automated Sorting:** Speeding up recycling and improving worker safety by instantly categorizing different waste types on conveyor belts.
- **Dynamic Route Optimization:** Reducing fuel use and emissions by directing garbage trucks only to areas that actually need pickup.
- **Illegal Dumping Detection:** Automatically spotting unauthorized trash in public spaces (like alleys or parks) for faster cleanup.